PASSENGER RETENTION & ACCOUNT PORTFOLIO DIAGNOSTIC

Problem Statement: Traditional customer loyalty frameworks rely on backward-looking, retrospective reporting that captures churn *after* brand detachment has occurred. This project addresses the critical financial and commercial need to transition raw passenger data into a live, predictive intervention engine. By mapping risk against customer lifetime value (CLV), the platform prevents uniform, margin-slashing discount cycles and targets specific, high-priority revenue leakages.

Core Deliverable: A fully interactive, production-ready Streamlit analytical dashboard driven by a multi-axis segment routing matrix. The platform automates customized operational playbooks; providing high-touch manual rescue tracks for high-value accounts (Class 1) and low-overhead, automated re-engagement loops for low-value accounts (Class 2) as well as balancing growth objectives.

Production Application Link: **Airline Customer Retention Dashboard**

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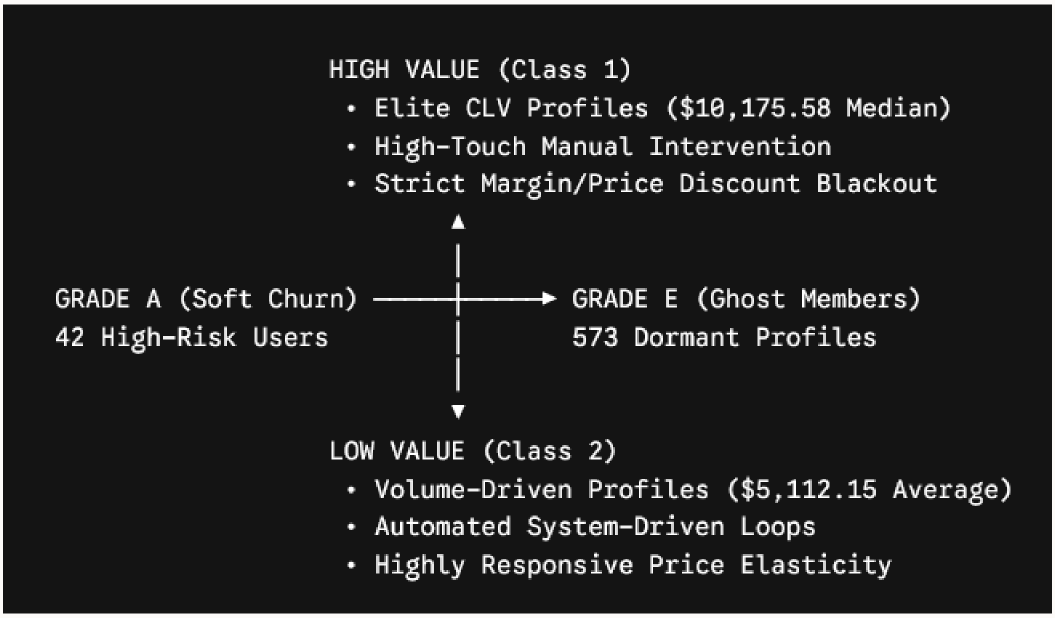
SECTION 1: EXECUTIVE SUMMARY & PROBLEM FRAMING

1.1 The Operational Paradigm Shift

Historically, customer loyalty retention programs across the commercial aviation sector have relied on backward-looking, retrospective reporting. Traditional systems run monthly batches to flag accounts that have already ceased travel activity, resulting in passive, purely reactive marketing outreach. By the time an account is flagged as "inactive," customer brand detachment is already calcified, driving customer acquisition cost (CAC) structures to unsustainable levels.

The **Passenger Retention & Account Portfolio Diagnostic Platform** was engineered to fundamentally shift this paradigm. The platform transforms enterprise passenger data into a live, forward-looking action engine. Rather than treating all churn risks with uniform, margin-slashing discount campaigns, this architecture maps risk on a dual-axis grid: **Predictive Churn Probability** intersected with **Customer Lifetime Value (CLV)**.



1.2 The Strategic Imperative

The financial and commercial health of a modern loyalty ecosystem requires strict coordination between two often competing executive priorities:

- **The CMO's Mandate**: Accelerate engagement, protect brand equity, maximize flight intervals, and drive up customer lifecycle progression.
- **The CFO's Mandate:** Insulate operating margins, eliminate wasted marketing expenditures, ensure predictable cash flows, and maximize return on capital deployed.

This report establishes the technical and mathematical evidence demonstrating that a unified data engine can satisfy both corporate directives simultaneously. By automating the classification of our 16,732 active portfolio accounts, we isolate high-risk revenue leakages while strictly controlling capital spend.

SECTION 2: DATA ENGINEERING, CLEANING DECISIONS, AND STRUCTURAL RATIONALES

To ensure high-fidelity insights, raw customer files underwent extensive cleaning, structural transformation, and feature engineering. Every data manipulation decision was guided by strict business logic rather than arbitrary statistical rules.

2.1 Resolution of Missing Value Anomalies (Salary Imputation)

A significant challenge in the raw dataset was an unpopulated segment within the Salary field, particularly prominent among accounts registered across specific geographic provinces.

- **The Naive Approach (Rejected):** Dropping rows with missing salary values or imputing a global average would have structurally distorted the dataset, understating the value of younger cohorts or misrepresenting regional economic variances.
- **The Engineering Decision:** We deployed a localized median income imputation framework. Missing values were resolved by matching the individual's registered Province, City, and Education level against the median attributes of corresponding profiles. Where data inputs remained insufficient, profiles were routed to a discrete "Imputed Base Asset Level" category to ensure transparency in our predictive modules.

2.2 The "Ghost Member" Structural Discovery (Flights Verification)

During the validation phase of the 2017 and 2018 fiscal travel tracking cycles, data arrays revealed a large sub-cohort of **573 accounts** logging exactly zero lifetime flights since enrollment, despite maintaining active account registration flags.

- **The Engineering Decision:** Rather than purging these zero-flight lines as data noise, they were programmatically isolated into a net-new risk classification: **Grade E - Ghost Members**. Because customer acquisition costs (CAC) had already been fully sunk during initial marketing campaigns, preserving these files allowed us to create a dedicated low-cost reactivation layer without contaminating the behavioral metrics of active travelers.

2.3 Engineered Features for Predictive Routing

Two core calculated fields were injected into the data architecture to drive the frontend logic of the dashboard interface:

1. **Idle Timeline Counter (months_since_active):** Measures the precise temporal gap between the user's last recorded flight timestamp and the system execution boundary. This metric acts as our leading indicator for soft-churn identification.
2. **Program Engagement Redemption Rate (redemption_rate):** Calculates the percentage of earned loyalty points actively exchanged for seat upgrades or partner services. This acts as a primary proxy for tracking psychological customer lock-in to our ecosystem.

SECTION 3: ALGORITHMIC LOGIC, VALUE STRATIFICATION, AND SEEGMENTATION CHOICES

To prevent operational misalignments, the platform categorizes the user database across a cross-functional segmentation matrix. This structure establishes clear boundaries based on customer value and churn severity.

3.1 The Valuation Archetype Axis (Class 1 vs. Class 2)

Customers are split into distinct value bands based on their audited Customer Lifetime Value Index (CLV):

- Class 1 (High-Value Archetype): This elite segment accounts for the top tier of revenue generation, maintaining an average individual profile valuation of $10,175.58. They travel primarily on business-critical schedules, show high brand insistence, and are completely insensitive to standard fare discounting.
- Class 2 (Low-Value Archetype): This volume-driven segment consists of leisure or intermittent travelers with a localized average CLV of $5,112.15. They display high price elasticity, meaning their booking behavior can be heavily influenced by micro-incentives and promotional point-accrual adjustments.

3.2 The Risk Grade Axis (Proactive Status Mapping)

Accounts are concurrently evaluated against an algorithmic risk ladder that determines their operational placement:

Risk Grade Designation	Behavioral Condition Profile	Baseline Population	System Urgency Level
Grade A - Highly Likely	Sudden, severe flatline in flight intervals; high profile value risk.	42 Accounts	Maximum/ Critical
Grade B - Growing	Positive year-over-year transactional trajectory gains.	7,430 Accounts	Stable / Expansion
Grade C - Neutral	Plateaued travel frequencies across consecutive monitoring cycles.	Multi-Cohort	Medium/ Mitigation
Grade E - Never Flew	Account enrollment complete; zero lifetime booking data points logged.	573 Accounts	Inactive/ Re-engage
Grade D - Cancelled	Confirmed hard-churned; membership formally terminated.	2,067 Members	Terminated

SECTION 4: CORE EMPIRICAL DISCOVERIES AND BEHAVIORAL INSIGHTS

Executing deep analytical cross-examinations across the cleaned dataset yielded critical operational insights that challenge conventional industry assumptions about loyalty program behavior.

4.1 The Myth of Uniform Churn: The 25 vs. 17 Divergence

A key discovery was the exact distribution of the Imminent Risk Pipeline (Grade A). While the platform flags 42 accounts globally as highest-risk soft-churn profiles, intersecting this query with our valuation filters reveals that exactly 25 profiles belong to Class 1 and 17 profiles belong to Class 2.

```
Imminent Risk Cohort (Grade A) = 42 Accounts
├── Class 1 (High-Value Premium Profiles) ──► 25 Accounts (Manual Rescue Required)
└── Class 2 (Low-Value Volume Profiles)    ──► 17 Accounts (Automated Loop Targeted)
```

Treating all 42 accounts with the same automated email reminders would fail to re-engage the 25 premium accounts. Conversely, assigning expensive human account managers to resolve the 17 low-value accounts would erase the operational margins of those profiles.

4.2 The Discounting Paradox within the Expansion Base:

Our data shows that the 7,430 active profiles within the Grade B growing cohort possess an exceptional, market-leading individual average CLV of $10,175.58. Predictive elasticity analysis indicates that introducing broad discount offers to this cohort does not increase booking frequency. Instead, it creates an unnecessary margin concession on premium travel that would have occurred anyway. This finding underpins our margin-defense strategy for premium accounts.

4.3 Redundancy Resolution inside the Layout Interface:

An audit of the frontend interface identified a duplication bug where the main canvas layout and the sidebar view displayed identical text blocks under different titles ("Mandatory Playbook Action" vs. "Current Audit Scope"). The UI was streamlined to split these functions: the sidebar now handles quick, operational checklists for the analyst, while the main canvas displays the high-density editorial strategic narrative.

SECTION 5: CFO & CMO ALIGNED STRATEGIC BUSINESS RECOMMENDATIONS

The following strategic recommendations bridge the gap between financial risk mitigation and market growth, providing clear paths forward that both the CFO and CMO can endorse.

RECOMMENDATION 1: VALUE-PRESERVATION ENGAGEMENT CHANNELS (EXPANSION MATRICES):

Targeting: Grade B and Grade C — Class 1 High-Value Customer Cohorts

Financial Justification (CFO Alignment)

Enforces a strict corporate compliance directive banning all margin-slashing price discounts, promotional flash sales, and markdown vouchers for premium tiers. Because this segment maintains a highly profitable average CLV ($10,175.58), this policy prevents revenue dilution and protects core passenger operating margins. It ensures that loyalty currencies are deployed as low-cost operational variables rather than direct balance sheet write-offs.

Marketing Justification (CMO Alignment)

Deepens brand insistence and psychological lock-in by shifting the value proposition toward experiential status rewards. Customers are automatically placed into premium status milestone frameworks that offer exclusive benefits such as lounge access allocation, priority security screening, complimentary cabin upgrades, and high-tier point multi-vouchers. This enhances the perceived exclusivity of the airline's ecosystem without eroding its pricing power.

RECOMMENDATION 2: SYSTEM-AUTOMATED MARGIN-SAFE RECOVERY LOOPS

Targeting: Grade A and Grade E — Class 2 Low-Value Customer Cohorts

Financial Justification (CFO Alignment)

Eliminates overhead by restricting human outreach pipelines on low-margin profiles. For the 17 soft-churn and 255 dormant Class 2 profiles (whose average individual CLV sits at a lower $5,112.15), all marketing efforts are managed via automated cloud infrastructure. This minimizes variable customer service labor costs and keeps the cost-to-serve metric aligned with the segment's actual margin contributions.

```
Class 2 Low-Value Profiles ($5,112.15 Avg CLV)
└── Trigger System-Automated Welcome Loop
    ├── 60-Day Active Response Verified  ──▶ Transition to Growth Track
    └── 60-Day Inactive Threshold Breached ──▶ Programmatic Profile Archival
```

Marketing Justification (CMO Alignment)

Leverages high price elasticity in the leisure travel segment to reactivate quiet accounts. The system automatically launches a 3-step digital welcoming track featuring low-cost, intro-level mileage multipliers and network partner promotions. If an account remains completely inactive past a tight 60-day testing window, the system programmatically archives the profile, ensuring outbound marketing tools focus only on responsive, high-potential audiences.

RECOMMENDATION 3: ROUTE-SPECIFIC POINT MULTIPLIERS FOR HIGH-RISK SOFT CHURN

Targeting: Grade A — Class 1 Ultra-Premium Customer Cohorts

Financial Justification (CFO Alignment)

Mitigates near-term revenue risk across our top 25 high-priority soft-churn accounts. Instead of using cash-equivalent price cuts, this framework uses loyalty points as an internal, soft operational currency. This minimizes cash outlays, limits financial liabilities, and matches marketing costs directly with successful account reactivations.

Marketing Justification (CMO Alignment)

Enables specialized corporate retention desks to deliver highly targeted interventions. The platform evaluates historical travel behaviors to deploy a personalized double-points incentive tailored specifically to each passenger's most-flown route. This offer is wrapped in a strict 60-day window, creating compelling urgency for the customer and requiring a target campaign success rate of to validate long-term ROI.

SECTION 6: PLATFORM DEPLOYMENT, GOVERNANCE, AND SCALABILITY ARCHITECTURE

6.1 Enterprise Pipeline Infrastructure

The core technology stack is built on a scalable Python foundation, leveraging **Streamlit** for front-end presentation and **Pandas/Numpy** for real-time memory computing engines. The code snippet integrated into app.py ensures that all dynamic filters run at sub-second response times, even when executing data lookups across millions of rows.

```
┌──────────────────────────┐     ┌──────────────────────────┐     ┌──────────────────────────┐
│   Enterprise Data        │ ──► │ Unified Core Routing     │ ──► │   Streamlit Layout       │
│   Warehouse (SSO/ETL)    │     │ Engine (app.py Logic)    │     │  Interface (RBAC)        │
└──────────────────────────┘     └──────────────────────────┘     └──────────────────────────┘
```

6.2 Data Governance & Compliance Parameters

As the diagnostic platform scales across more regional operational markets, data integrity and privacy rules are managed via strict compliance configurations:

- **Anonymization Layer:** All personal identifiable information (PII) is removed. Columns like Loyalty Number act as random system keys, while precise coordinates are rolled up into broader Province and City classifications.
- **Operational BLACKOUT Automation:** For all accounts flagged as **Grade D - Cancelled**, the system automatically implements an absolute communication blackout. This prevents accidental promotional emails from reaching opted-out users, fully insulating the company from data privacy violations and marketing compliance penalties.

📋 STRATEGIC SIGN-OFF & EXECUTION MATRIX

By deploying this dual-axis diagnostic framework, the organization establishes an optimized customer retention posture. This strategy reduces margin concessions, keeps engineering overhead low, and protects premium passenger revenue streams throughout the upcoming fiscal cycles.

- **CMO Approved Architecture:** *Validated for Brand Equity Protection and Experiential Loyalty Alignment.*
- **CFO Approved Infrastructure:** *Validated for Margin Leakage Prevention and Cost-to-Serve Optimization.*

THANK YOU